



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BT	Semester:	V
Course Code:		Course Name:	AI

Name of Student:	SAHIL ANIL VINOD
Roll No. :	61
Assignment No.:	1
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	04
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	08

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Ms. Sneha Yadav

Signature :

Date :

Q 1) A delivery drone learns the model shortest path to its destination without knowing the environmental model. find how Q-learning updates q-values to improve the drone's policy.

→ when a delivery drone doesn't know the environmental model, it uses Q-learning a model-free reinforcement learning algorithm, to learn shortest path by trial and error.

Core idea of Q-learning -

$$Q(s, a)$$

∴ s = current state

∴ a = action

∴ $Q(s, a)$ = expected future reward of taking action in state

$$Q(s, a) = Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

where,

α (alpha) = learning rate

γ (gamma) = discount factor

r = reward received

s' = next state

$\max Q(s', a')$ = best possible future value

How the drone learns the shortest path -

S1 - Initialization

- Q-table initialization with zeros
- drone starts exploring randomly.

S2 - Take action -

from state s , drone chooses action -

- Random (exploration)

- Best Q-value (exploitation), called ϵ -greedy policy.

S3 - Received reward -
situation

	Reward
reached destination	-
hit obstacle	+100
normal move	-100
	-1

S4 - Update Q-value

Suppose - $Q(s, a) = 5$

$$\alpha = 0.1$$

$$\gamma = 0.9$$

$$r = -1$$

$$\max_{a'} Q(s', a') = 8$$

update -

$$Q = 5 + 0.1 [(-1 + 0.9(8) - 5)]$$

$$Q = 5 + 0.1 [(-1 + 7.2 - 5)]$$

$$Q = 5 + 0.1(1.2)$$

$$Q = 5.12$$

why this finds the shortest path -

every move costs -1 reward

longer paths - more negative total reward

shortest path - highest total reward

How policy improves over time -

Initially - agent explores randomly

- Q-values inaccurate

- After many episodes - wrong paths $\rightarrow$ low Q-values

- efficient paths $\rightarrow$ high Q-values

- agent follows

$$\pi(s) = \arg \max_a Q(s, a)$$

FOR EDUCATIONAL USE

Q. 2) A game receives +10 for winning, -1 for each move, and -50 for losing the game. explain how this reward design influences the agent's learning behaviour.

-
- +10 → winning
 - 1 → each move
 - 50 → losing

∴ The agent tries to maximize total cumulative reward:

$$G_t = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \dots$$

How each reward influences behavior -

① +10 for winning → encourages goal achievement

- winning gives positive reinforcement
- the agent learns which state-action sequences lead to victory
- paths leading to win get higher value estimates

② -1 per move → encourages efficiency

- This is called a step penalty.

effects - longer games → more negative reward accumulated.

- shorter path to win → higher total return.

Eg. - win in 5 moves -

$$+10 - 5 = +5$$

- win in 15 moves -

$$+10 - 15 = -5$$

③ -50 for losing → strong loss avoidance

- This large negative reward creates -

- high penalty for risky strategies.
- strong discouragement for moves leading to losing state.

Eg. lose in 5 moves.

$$\therefore -50 - 5 = -55$$

- This is much worse than any slow in.

∴ The agent becomes - Risk aware

- deference

- careful near dangerous states

• combined behavioral effect -

The agent learns to - win, win quickly, avoid losing at cost.

- This creates a balanced strategy.

- Aggressive enough to win

- efficient in action selection

- careful about risky decisions.

• Trade-off analysis -

if -50 is too large -

- Agent may become overly cautious.

- might prefer safe draw-like behavior

- if -1 per move is removed -

- agent might take unnecessarily long routes.

- if +10 is too small -

- agent may focus more on avoiding loss than trying to win.

• learning outcome over time -

eventually, the policy becomes -

$$\pi(a) = \argmax_a Q(s, a)$$

- It encourages fast, safe, and winning strategies.